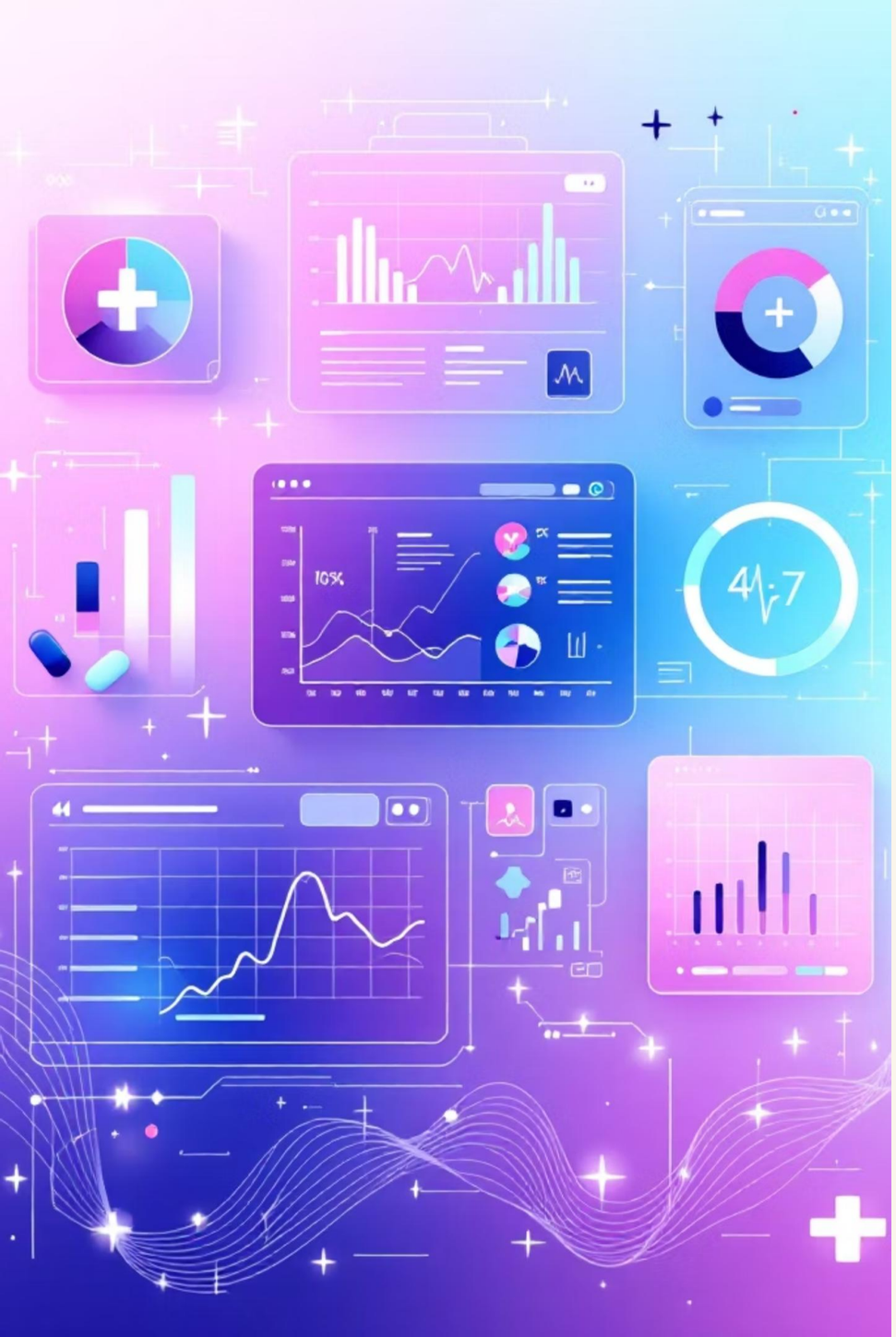

InsurePro: AI-Based Insurance Premium Prediction System

By Kanak Raj Arora

Under Guidance of Dr. Amandeep Cheema

Department of CSE

Manipal University Jaipur

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Problem Statement

Insurance companies struggle to accurately predict medical premiums.

Traditional actuarial models:

- Assume linear relationships
- Fail to capture complex interactions
- Lead to inaccurate pricing

Objective:

Build an accurate, interpretable, and deployable ML-based system

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Research Context

Challenges

- Rising healthcare costs
- Complex non-linear relationships
- Need for transparent decision-making

Opportunity

Machine Learning for better risk-based pricing

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Research Gaps Addressed

Stacking Ensemble

Limited implementation of stacking approaches combining multiple heterogeneous base learners for medical insurance prediction.

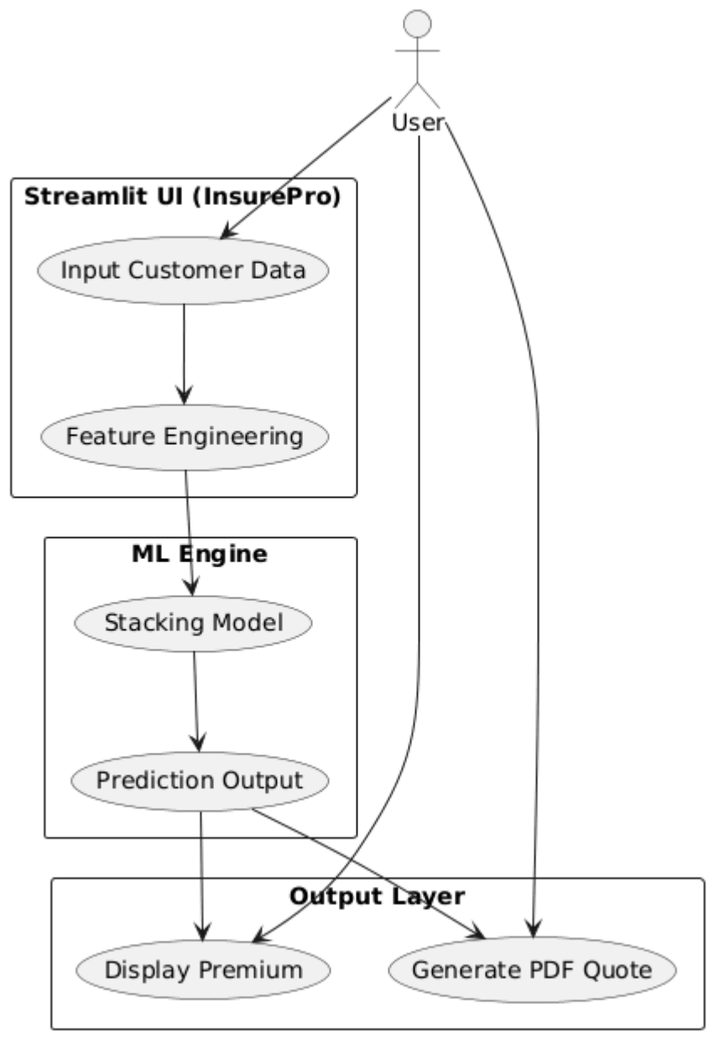
End-to-End System

Need for integrated frameworks combining data preprocessing, model training, prediction, and report generation.

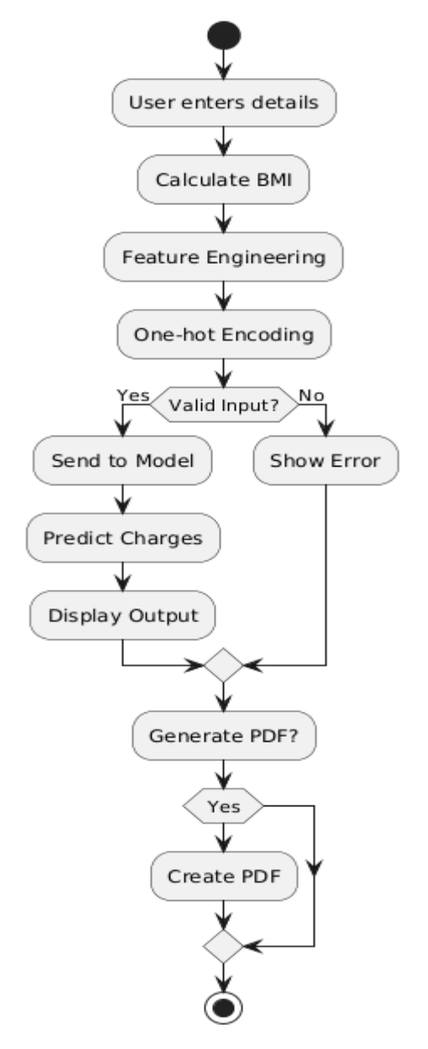
Feature Interactions

Complex interactions between demographic and health-related features require sophisticated feature engineering.

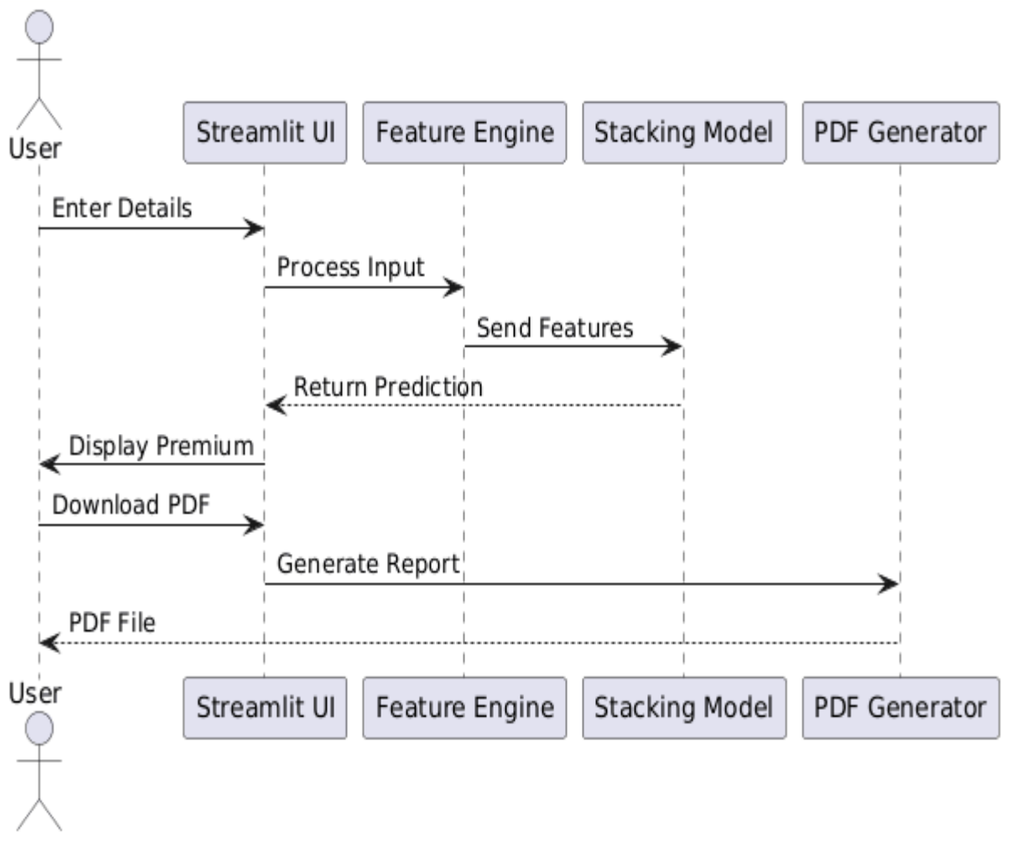
Diagrams



Architecture Diagram



Activity Diagram



Sequence Diagram

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Methodology Overview

01

Dataset

2,772 insurance records with 7 features: age, sex, BMI, children, smoking status, region, and charges.

02

Preprocessing

- Duplicate removal
- Missing values removal

03

Feature Engineering

BMI categories, age groups, and interaction features created to enhance model performance.

BMI Categories

- *Underweight (<18.5)*
- *Normal (18.5–25)*
- *Overweight (25–30)*
- *Obese (>30)*

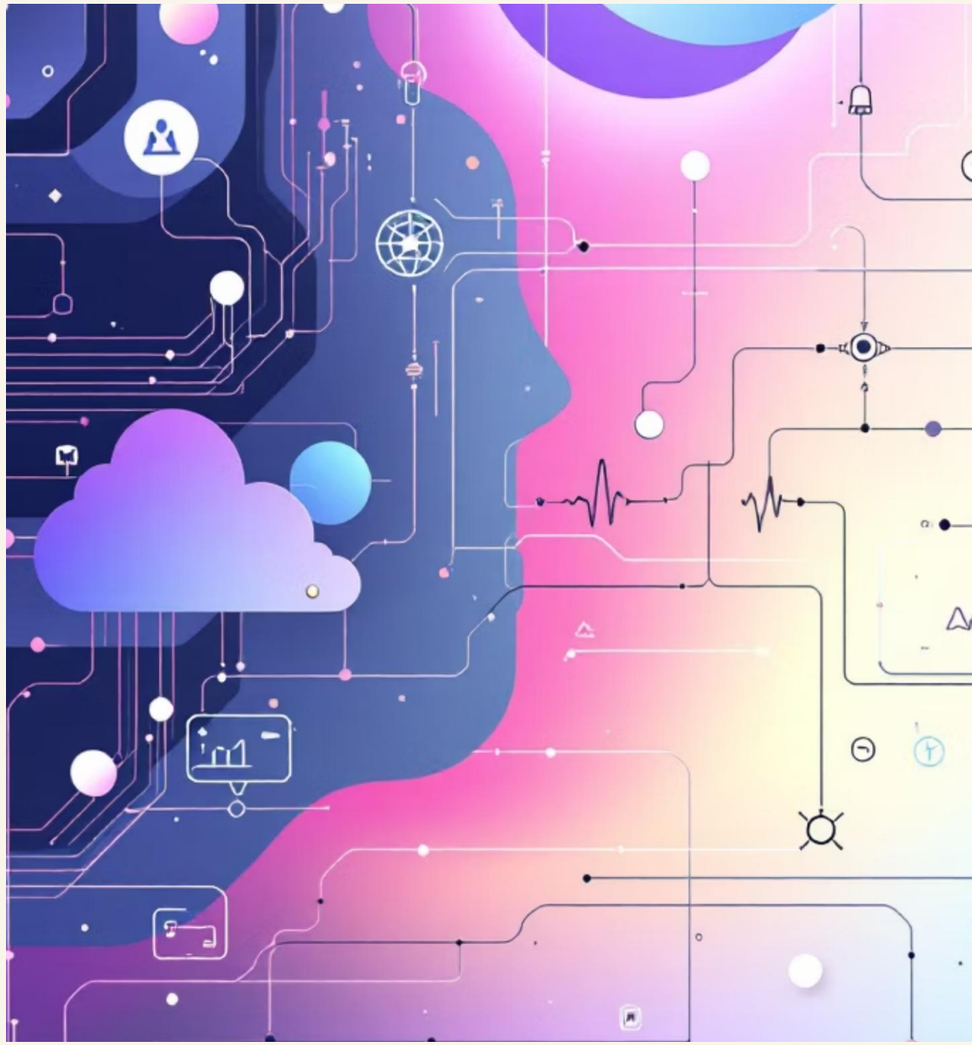
Age Groups

- *Teen (18–25 years)*
- *Young Adult (26–35 years)*
- *Adult (36–50 years)*
- *Senior (51–65 years)*

04

Model Development

Stacking ensemble combining Random Forest, Gradient Boosting, and XGBoost with Ridge meta-learner.



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Model Architecture

Base Models

Random Forest
- Bagging-based ensemble
- 200 trees, max depth = 5
- Handles variance and overfitting

Gradient Boosting
- Sequential error correction
- 300 estimators, learning rate = 0.01
- Captures complex non-linear patterns

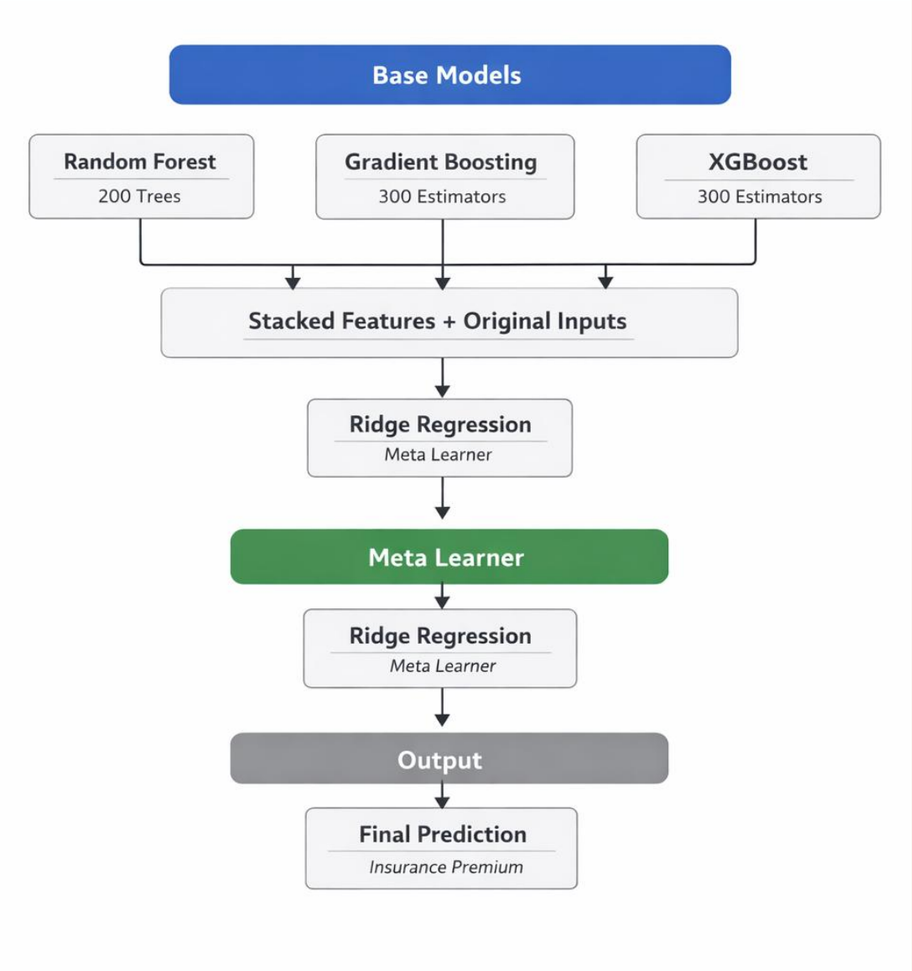
XGBoost
- Optimized gradient boosting
- 300 estimators, max depth = 3
- Regularization + high efficiency

Meta Learner

Ridge Regression
- Combines predictions from base models
- Uses regularization (λ = 10)
- Final prediction generation

Final Output

Stacking Ensemble Model
- Integrates base + meta learning
- Improves accuracy and generalization



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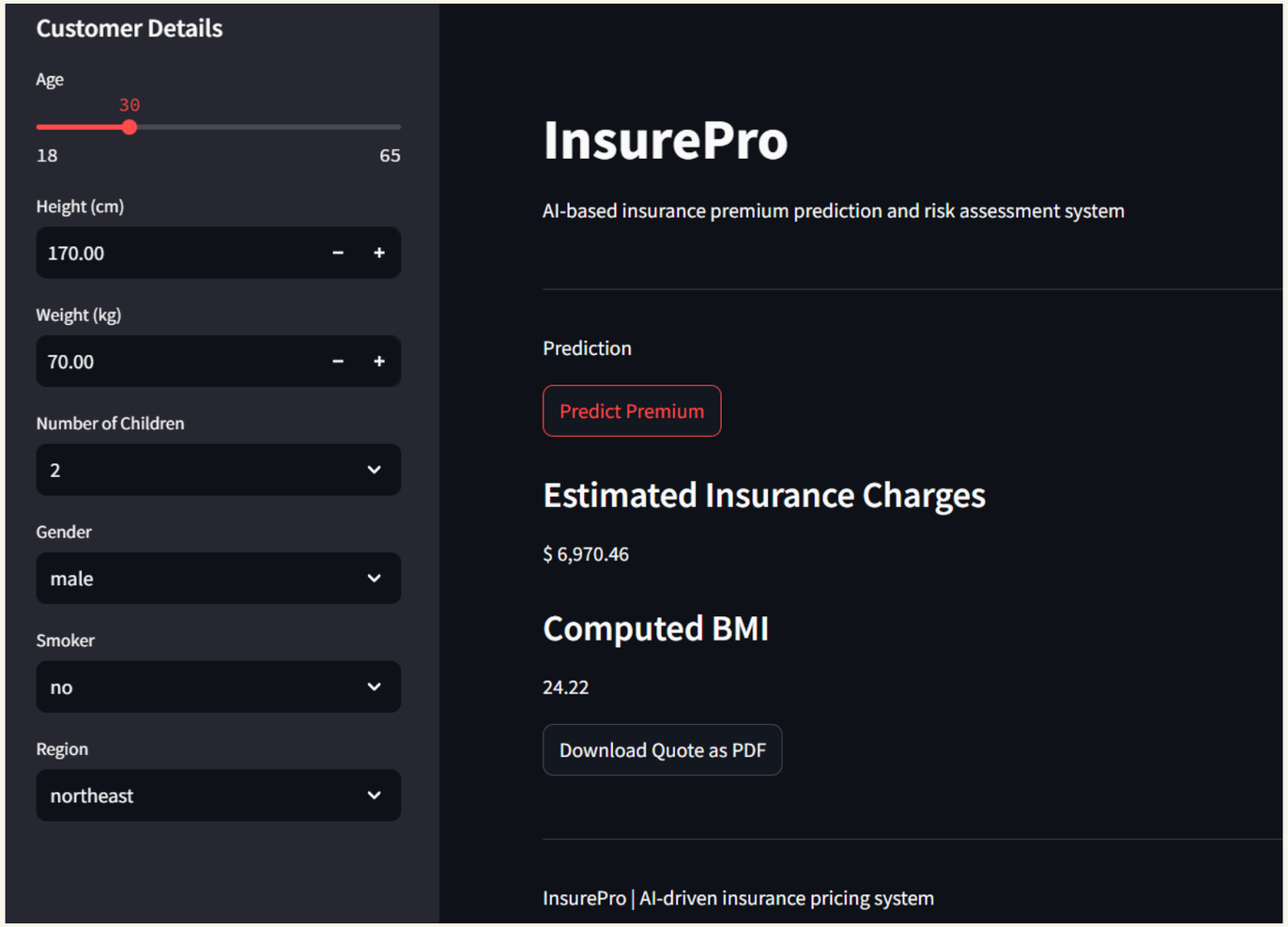
Results Comparison

Model	R^2	RMSE	MAE
Dummy Regressor (Baseline)	-0.0084	13612.43	9861.80
Linear Regression	0.889	4524.85	2784.31
Ridge Regression	0.889	4509.62	2781.12
Lasso Regression	0.889	4524.50	2784.32
Decision Tree	0.891	4471.10	2684.19
Random Forest	0.880	4692.85	2656.41
Gradient Boosting	0.901	4273.12	2463.78
AdaBoost	0.837	5470.78	4719.83
XGBoost	0.887	4565.93	2490.08
Stacking (RF+GB+XGB)	0.902	4248.75	2449.45
Stacking with Ridge	**0.905**	**4183.14**	**2419.77**

The stacking ensemble achieved 0.4% improvement over the best individual model (Gradient Boosting) and dramatically outperformed the baseline.

System Demo



Features:
- User inputs (age, height, weight)
- BMI calculated dynamically
- Real-time premium prediction
- Risk classification
- PDF quote generation

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Future Directions

Deep Learning Integration

Hybrid approaches combining stacking ensembles with deep neural networks for complex pattern recognition.

Expanded Features

Incorporate medical history, pre-existing conditions, family health history, and lifestyle factors.

Cloud Deployment

Deploy on cloud platforms with auto-scaling for large-scale production workloads.

Fairness & Bias

Fairness-aware ML techniques to ensure equitable premium pricing across demographic groups.

Conclusion

Superior Performance

Stacking ensemble achieved R^2 of 0.905, RMSE of 4,183, and MAE of 2,420, outperforming all individual models.

Feature Engineering Impact

BMI categories, age groups, and interaction features significantly improved accuracy over regression baselines.

Ensemble Effectiveness

Ensemble methods effectively captured complex non-linear relationships in healthcare cost data.

Limitations

- Small dataset (2772 records)
- Limited features (no medical history)
- Slight prediction variance in high-cost cases

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Thank You